Cambridge Assessment International Education

Cambridge International AS & A Level

BIOLOGY

9700/13

Paper 4 A Level Structured Questions – Investigation of Limiting

MARK SCHEME

Maximum Mark: 131

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

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This document consists of 13 printed pages.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:
• the specific content of the mark scheme or the generic level descriptors for the question
• the specific skills defined in the mark scheme or in the generic level descriptors for the question
• the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always whole marks (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded positively:
• marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
• marks are awarded when candidates clearly demonstrate what they know and can do
• marks are not deducted for errors or omissions
• answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question.

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles & Abbreviations

1 Context and scientific use of keywords: marks should not be awarded if keywords are used incorrectly.
2 Contradictory statements: credit should not be awarded for any correct statement that is contradicted within the same question part.
3 Calculation guidance: correct answers to calculations should be given full credit even if there is no working, unless stated otherwise.
Abbreviations: ; separates marking points | **/** alternative answers | **R** reject | **A** accept | **AW** alternative wording | **underline** essential word | **max** maximum mark | **ecf** error carried forward | **I** ignore | **AVP** alternative valid point.

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Question	Answer	Marks
1(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	**2**
1(b)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	**2**
1(c)	• correct biological term / structure identified ; ***R*** *ambiguous terminology*	**1**
1(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	**3**
2(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	**1**
2(b)	• correct biological term / structure identified ; ***R*** *ambiguous terminology*	**1**

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Question	Answer	Marks
2(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
2(d)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
3(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
3(b)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	1
3(c)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	1

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Question	Answer	Marks
3(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	3
4(a)	• correct biological term / structure identified ; • further elaboration of biological principle / detail of mechanism ; ***R*** *ambiguous terminology*	2
4(b)	• detailed account of sequence of events / structural features observed ; • specific factual details / trends / data points cited ; ***A*** *descriptive points in any logical order*	2
4(c)	• correct biological term / structure identified ; ***R*** *ambiguous terminology*	1
4(d)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
5(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
5(b)	• correct biological term / structure identified ; ***R*** *ambiguous terminology*	1

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Question	Answer	Marks
5(c)	• accurate biological statement meeting the syllabus learning objective ; ***A*** *acceptable syllabus alternative [AW]*	1
5(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
1(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
1(b)	• accurate biological statement meeting the syllabus learning objective ; • supporting detail / elaboration of biological concept ; • further elaboration of biological principle / detail of mechanism ; ***A*** *acceptable syllabus alternative [AW]*	3
1(c)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
1(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	3

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Question	Answer	Marks
2(a)	• formula: magnification = image size / actual size / $M = I / A$ (or $A = I / M$) ; • correct conversion of units (e.g. mm × 1 000 = µm / cm × 10 000 = µm) ; • correct final answer with appropriate units (µm / nm) or '×' for magnification ; ***A*** *correct numerical answer alone scores full marks* ***ecf*** *error carried forward if formula is correct but conversion error occurred*	1
2(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
2(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	5
2(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
3(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2

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Question	Answer	Marks
3(b)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	3
3(c)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	3
3(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
4(a)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	7

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Question	Answer	Marks
5(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; • further elaboration of biological principle / detail of mechanism ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	**8**
1(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	**2**
1(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**3**
1(c)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**3**
1(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	**2**

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Question	Answer	Marks
2(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	3
2(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; • further elaboration of biological principle / detail of mechanism ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	4
2(c)	• glycolysis in cytoplasm: glucose phosphorylated using ATP to fructose-1,6-bisphosphate ; • split into two triose phosphate (TP), oxidized to pyruvate producing net 2 ATP and 2 reduced NAD ; • link reaction: pyruvate decarboxylated and dehydrogenated to acetyl CoA in mitochondrial matrix ; • Krebs cycle: acetyl CoA combines with oxaloacetate to form citrate; cyclic reactions yield ATP, CO2, reduced NAD, reduced FAD ; • chemiosmosis: protons pumped into intermembrane space, flow through ATP synthase generating ATP ; ***A*** *balanced equation: C6H12O6 + 6O2 -> 6CO2 + 6H2O*	2
2(d)	• glycolysis in cytoplasm: glucose phosphorylated using ATP to fructose-1,6-bisphosphate ; • split into two triose phosphate (TP), oxidized to pyruvate producing net 2 ATP and 2 reduced NAD ; • link reaction: pyruvate decarboxylated and dehydrogenated to acetyl CoA in mitochondrial matrix ; • Krebs cycle: acetyl CoA combines with oxaloacetate to form citrate; cyclic reactions yield ATP, CO2, reduced NAD, reduced FAD ; • chemiosmosis: protons pumped into intermembrane space, flow through ATP synthase generating ATP ; ***A*** *balanced equation: C6H12O6 + 6O2 -> 6CO2 + 6H2O*	2
3(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	1

Question	Answer	Marks
3(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	2
3(c)	• detailed account of sequence of events / structural features observed ; • specific factual details / trends / data points cited ; • comprehensive description addressing all aspects of the process / observation ; • further elaboration of biological principle / detail of mechanism ; ***A*** *descriptive points in any logical order*	4
3(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
4(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	1
4(b)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; • detailed biochemical context: reference to specific molecular interactions / syllabus concepts ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	3

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Question	Answer	Marks
4(c)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	3
4(d)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; • further elaboration of biological principle / detail of mechanism ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	6
5(a)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
5(b)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	2
5(c)	• light-dependent stage: on thylakoid membranes, photolysis of water releases H+, e-, and O2 ; • non-cyclic photophosphorylation generates ATP and reduced NADP ; • light-independent stage / Calvin cycle in stroma: CO2 fixed by RuBisCO with RuBP forming GP ; • GP reduced to triose phosphate (TP) using ATP and reduced NADP ; • DCPIP acts as an artificial electron acceptor, reduced from blue to colourless by electrons from photosystems ; ***A*** *limiting factors: light intensity, CO2 concentration, temperature*	3

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Question	Answer	Marks
5(d)	• biological mechanism: relevant physiological / molecular process clearly outlined ; • cause and effect linked: explanation of how structure or condition relates to function ; ***A*** *scientifically equivalent wording [AW]* ***I*** *vague references without specific biological terms*	**2**

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